AppliedAI Report

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1 Introduction

Lung cancer remains the leading cause of cancer-related mortality worldwide, accounting for approximately 18% of all cancer deaths in 2020. Its high fatality rate is primarily attributed to late-stage diagnoses, which significantly reduce the chances of successful treatment. Early and accurate detection of malignant nodules through screening programs is, therefore, critical to improving patient outcomes.

Recent advances in Artificial Intelligence (AI) and deep learning have opened new avenues for assisting radiologists in the interpretation of medical imaging. In particular, Convolutional Neural Networks (CNNs) have shown promising capabilities in classifying and detecting malignancies from CT scans. These methods can potentially increase diagnostic accuracy and reduce inter-observer variability, thereby supporting more consistent and timely clinical decisions.

This project addresses the task of lung nodule classification using CT scan data. Each patient in the dataset is represented by two CT image slices—one full slice and one zoomed-in on the nodule—with associated malignancy scores ranging from 1 (least malignant) to 5 (most malignant). The goal is to develop AI models that can classify these images in two different ways: (i) a binary classification into benign (classes 1–3) vs. malignant (classes 4–5), and (ii) a five-class malignancy scoring.

By designing and evaluating deep learning models on both full-slice and zoomed-slice inputs, this project aims to assess the effectiveness of AI-based diagnostic tools in lung cancer detection and explore the impact of different input granularities on classification performance.

2 Material and methods

2.1 Dataset

The dataset used in this project consists of computed tomography (CT) scan slices collected from approximately 2,300 patients. Each patient is associated with two image slices:

- A full-slice image, which is a complete 512×512 scan of the thoracic region.
- A zoomed-in nodule image, which focuses on the region surrounding the largest detected lung nodule. These zoomed images vary in shape and size, depending on the original nodule dimensions and positioning.

Each sample is labeled with a malignancy score ranging from 1 to 5. This score serves as the ground truth for two classification tasks: a binary classification (benign: scores 1–3 vs. malignant: scores 4–5) and a multiclass classification (scores 1 through 5).

2.1.1 Preprocessing

To ensure compatibility with Keras-based convolutional neural networks, all images were rescaled to a fixed input size. Since the zoomed-in nodule slices were non-square and varied in shape, we applied a combination of interpolation and padding to standardize their dimensions while preserving the anatomical context. The target size was determined through preliminary analysis, taking into account both the visual resolution required to retain diagnostic features and the input size constraints of the selected model backbones.

CT image values, expressed in Hounsfield Units (HU), were filtered to retain only those within the range [-1000, +1500]. Element outside this interval were clipped to -1000, corresponding to air. This step aimed to remove extraneous visual noise and ensure that only relevant tissue regions were included, as human tissues generally fall within this interval.

Figure 1: Example of full-slice image after preprocessing.

2.1.2 Class Imbalance

A significant challenge emerged from the class imbalance in the dataset. Specifically, class 3 (moderate malignancy) appeared approximately five times more frequently than class 5 (most malignant), heavily skewing the class distribution. This imbalance adversely affected model training, especially in the five-class classification tasks, where early models converged on predicting only the dominant class.

Several strategies were explored to mitigate this issue:

- Class weighting in the loss function.
- Manual oversampling and downsampling of underrepresented and overrepresented classes.
- A hybrid of the two approaches.

While these methods showed some improvements, they were insufficient in isolation. Therefore, a custom Python class, BalancedDataGenerator, was implemented. This generator dynamically constructs training batches for each epoch by randomly sampling from the dataset while enforcing per-epoch class balance. This approach retained all available samples across training epochs without introducing persistent duplication or data loss.

2.1.3 Data Augmentation

To enhance generalization and reduce overfitting, on-the-fly data augmentation was applied during training. The augmentations included:

- Geometric transformations: random flipping, rotation, and translation (within a few pixels).
- Photometric changes: brightness and contrast adjustments.
- Noise injection: adding low-level random noise to simulate real-world variability.

These augmentations were recomputed randomly for each batch, resulting in slightly longer training times but ensuring a virtually infinite stream of unique image variations, thereby strengthening model robustness.

2.2 Model

For all four classification tasks, we adopted a transfer learning and fine-tuning strategy rather than training convolutional neural networks (CNNs) from scratch. This approach leverages pre-trained feature extraction capabilities while allowing flexibility in adapting to the domain-specific characteristics of medical imaging.

The backbone networks were initialized with ImageNet pre-trained weights. Despite the domain difference between ImageNet (natural images) and CT scan data, the lower layers of deep CNNs are known to learn generic features such as edges, shapes, and textures that are transferable and valuable for medical image analysis. To adapt the models for our tasks, the original classification heads were removed and replaced with task-specific output layers.

2.2.1 Nodules models

For the classification tasks based on zoomed-in nodule images, both the binary (benign vs. malignant) and the five-class malignancy models were built using a DenseNet backbone. This architecture was selected after evaluating several alternatives, as it consistently delivered better results on the localized nodule regions.

Each model architecture consists of:

1. A DenseNet feature extractor pre-trained on ImageNet.
2. A Global Average Pooling layer to reduce spatial dimensions and introduce translation invariance.
3. A Dense hidden layer with 100 units, adding learning capacity and enabling the model to capture complex patterns.
4. A Dense output layer:
   - With 2 units and softmax activation for the binary classification task.
   - With 5 units and softmax activation for the five-class task.

In the early stages of development, we experimented with architectures similar to those used for the full-slice models. However, we observed that the nodule images could benefit from a deeper architecture. while the full-slice models were more prone to overfitting, and thus performed better with a simpler output structure.

2.2.2 Full-slice models

For the classification tasks on full-slice CT images, we used a different backbones based on EfficientNetV2-S: Both models were initialized with ImageNet weights and share a common structure: a Global Average Pooling layer followed by a single Dense output layer with 2 units for the binary task and 5 units for the multiclass task, both using softmax activation.

This simple setup proved effective, as adding extra dense layers increased the risk of overfitting without improving accuracy on the validation set.

2.3 Evaluation method

To assess the performance of the models across the four classification tasks, we adopted a set of well-established evaluation metrics: Accuracy, Precision, Recall, and F1-score. These metrics provide a comprehensive overview of a model's behavior, especially in medical diagnostic tasks where class imbalance and misclassifications costs are critical considerations.

- Accuracy measures the overall proportion of correctly classified instances among all predictions.
- Precision (also known as Positive Predictive Value) indicates the proportion of true positive predictions among all positive predictions made by the model.
- Recall (or Sensitivity) represents the proportion of true positive predictions among all actual positives in the dataset.
- F1-score is the harmonic mean of Precision and Recall, providing a balanced metric when there is an uneven class distribution.

For the multiclass classification tasks, we used the macro-averaged versions of Precision, Recall, and F1-score. This approach calculates the metric independently for each class and then averages them, giving equal weight to each class regardless of its frequency. This is especially important given the class imbalance in the dataset, where dominant classes could otherwise skew the results.

2.3.1 Cross-Validation

Before evaluating the models on the final dedicated test set, we employed 5-fold cross-validation for model validation and hyperparameter tuning. In this setup, the dataset is randomly split into five subsets; in each round, four folds are used for training and one for validation, ensuring that every sample is used for validation exactly once. The reported cross-validation results represent the average of the performance metrics across all five folds.

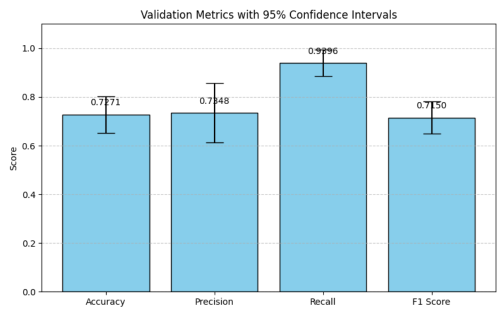

Figure 2: Example of the confidence intervals calculated for the binary full-slice model.

To provide a statistical understanding of model reliability, we also computed 95% confidence intervals for each metric using the scipy library (figure 2). These intervals offer an estimate of the variability and robustness of the model's performance across different data splits.

This combination of multiple metrics, macro-averaging, and confidence intervals ensures a balanced and rigorous evaluation framework suitable for both binary and multiclass diagnostic tasks.

3 Results

To evaluate the final performance of our models, we reserved 15% of the original dataset as a held-out test set, generated using stratified sampling to preserve the class distribution of the full dataset. In accordance with best practices, the test set was not used during model development or selection. All design decisions, hyperparameter tuning, and architecture choices were made based on performance on the training and validation sets only.

The final evaluation was conducted exclusively on the test set, and the results for each model across all four classification tasks are summarized in Table 2 that are consistent with the intervals found with cross-validation (Table 1).

When selecting the best-performing model for each task, we considered all four metrics, rather than relying solely on accuracy. This was a deliberate design choice intended to ensure a more balanced and robust evaluation. For example, a model with high accuracy may still perform poorly on minority classes if it fails to recall them appropriately: something not reflected in the accuracy metric alone.

However, we acknowledge that this strategy is arbitrary and application-agnostic. In a real-world deployment, the choice of the optimal model should be informed by domain expertise or by the specific requirements of the clinical or operational context. For instance, in a diagnostic setting, it may be more acceptable to misclassify a benign nodule as malignant (leading to a follow-up examination) than to miss a malignant tumor. In such cases, precision for the malignant class would be the most critical metric, and model selection should prioritize it accordingly.

4 Discussion and conclusion

To better understand the limited performance of the multiclass models, we employed two analysis methods: the confusion matrix, to examine the relationship between predicted and true labels, and the Class Activation Mapping (CAM) technique, to gain insight into the model's focus areas during classification.

4.1 Confusion matrix

The confusion matrix in Figure 3 provides a detailed view of how the model's predictions align with the true labels. As expected from the moderate accuracy, the results are not optimal; however, meaningful patterns emerge. Notably, the upper-right and lower-left corners of the matrix contain mostly zeros, indicating that the model rarely confuses the most dissimilar classes (e.g., it almost never classifies a highly malignant case as benign, or vice versa).

Additionally, a concentration of values around the diagonal suggests that the model tends to misclassify samples into neighboring classes, rather than random or unrelated ones. This indicates that the model has

Model	Accuracy	Precision	Recall	F1-Score
Full-slice Binary (EfficientNetV2-S)	0.72 ±0.07	0.73 ±0.12	0.93 ±0.05	0.71 ±0.06
Full-slice Multiclass (EfficientNetV2-S)	0.26 ±0.10	0.28 ±0.06	0.33 ±0.07	0.25 ±0.09
Nodule Binary (DenseNet)	0.86 ±0.02	0.82 ±0.05	0.66 ±0.08	0.63 ±0.02
Nodule Multiclass (DenseNet)	0.43 ±0.04	0.42 ±0.02	0.46 ±0.04	0.41 ±0.03

Table 1: Evaluation metrics for all models after cross-validation with confidence intervals.

Model	Accuracy	Precision	Recall	F1-Score
Full-slice Binary (EfficientNetV2-S)	0.67	0.66	0.39	0.73
Full-slice Multiclass (EfficientNetV2-S)	0.25	0.26	0.29	0.28
Nodule Binary (DenseNet)	0.83	0.69	0.60	0.64
Nodule Multiclass (DenseNet)	0.55	0.50	0.42	0.42

Table 2: Evaluation metrics for all models on the test set.

learned some useful structure in the data, even if its predictions are not fully accurate.

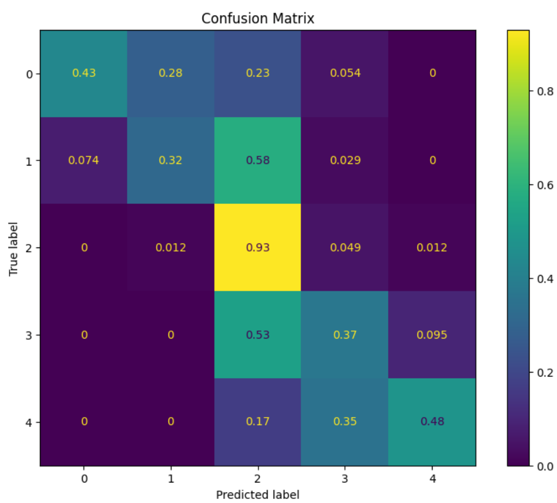

Figure 3: Confusion matrix calculated on the test set for the multiclass nodule model.

Notably, due to the high class imbalance favoring class 3, the model achieves high precision in identifying that class. However, this comes at the cost of frequently misclassifying instances of class 2 and class 4 as class 3, reflecting a bias toward the dominant class.

4.2 Explainability

One of the key challenges in using deep learning for medical applications is the lack of interpretability: even when models perform well, it is difficult to understand why they make certain predictions. To address this, we used Class Activation Mapping (CAM) to visualize the regions of the input that the model considers important.

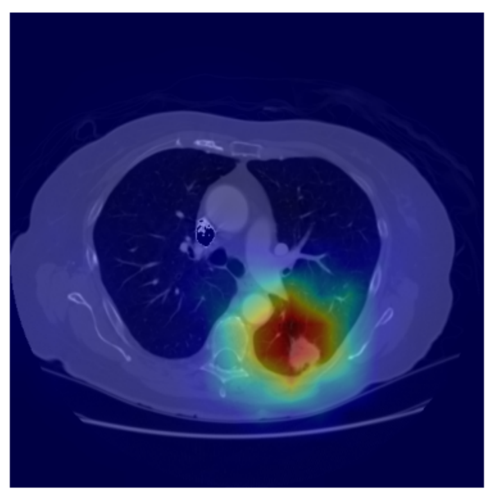

Figure 4: CAM heatmap overlapped with the input image for a class 5 tumor.

Given the relatively poor performance of the full-slice multiclass classifier, we applied CAM to investigate whether the model focuses on the relevant anatomical regions. As shown in Figure 4, the model generally identifies the correct area of interest. However, the activation regions are broad, suggesting that the model may be incorporating irrelevant contextual information along with the nodule. This could explain the lower classification performance, as excessive focus on non-discriminative features may dilute the model's decision-making process.

4.3 Model comparison

As expected, the complexity of the classification task and the nature of the input images significantly influenced model performance. Binary classification tasks consistently outperformed their multiclass counterparts, reflecting the increased difficulty of learning fine-grained distinctions between five malignancy levels as opposed to a simpler benign-versus-malignant dichotomy.

Moreover, models trained on cropped nodule images achieved higher accuracy and robustness than those trained on full-slice images. This is particularly evident in the binary classification task, where the DenseNet model trained on nodule crops achieved the highest overall performance, with an accuracy of 0.83 and an F1-score of 0.64 on the test set.

In the multiclass scenario, performance dropped considerably across all models. The full-slice multiclass classifier in particular performed poorly, with a test accuracy of only 0.25 and an F1-score of 0.28; those values are too low to be considered acceptable for clinical applications. The multiclass model based on nodule crops, while still limited, performed better (accuracy: 0.55; F1-score: 0.42), demonstrating the benefit of focusing on the region of interest.

These findings confirm the intuition that classification is more accurate when focused directly on the nodule rather than on the entire anatomical context, which may introduce noise and irrelevant information. They also reinforce the observation that binary classification is inherently more tractable than multiclass prediction, especially in the presence of class imbalance and ambiguous inter-class boundaries.